

FULL NAME: \_\_\_\_\_

BLOCK: \_\_\_\_\_

## Solving Radical Equations — Foundation

Lesson 5-4 · Period 1 · Teacher Edition

### OBJECTIVES

|                                |                                                                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Math Objective</b>          | Solve radical equations by isolating the radical and raising both sides to the appropriate power; identify and reject extraneous solutions.            |
| <b>Language Objective</b>      | Explain in writing, using sentence frames, why squaring both sides can introduce extraneous solutions and how to check.                                |
| <b>Essential Understanding</b> | Squaring both sides is NOT a reversible operation — it can create solutions that don't satisfy the original equation. Every candidate must be checked. |

### TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

|                           |                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Topic Goals</b>        | Solve single-radical equations; detect extraneous solutions.                                        |
| <b>Essential Question</b> | Why does squaring both sides sometimes produce solutions that don't work, and how do we catch them? |
| <b>Materials</b>          | Student packet, pencil. Teacher models Example 1, then gradual-release into Example 3 (extraneous). |

### [PIN] PRE-FLIGHT NOTE (read before class)

Explore is 6 bank items: Try-It 1 → Try-It 3 → #22 → #8 → #15 → #4. Paced for ~5–6 min per item.

If pairs finish early, push them to the Optional Reinforcement: Model & Discuss Parts B + C (verbal/TPS), then Practice #21.

If pace slows, cut #4 (extension) — it's the DOK-2 stretch; the core DOK-1 arc (Try-It 1 → Try-It 3 → #22 → #8 → #15) is the minimum viable set.

*Tag: notice & wonder*

#### Do Now

DOK: 2

Minutes: 5

#### Teacher does

- Project the starter (or use printed Do Now sheet).
- Circulate silently. Do NOT teach during this phase.
- Collect at 5 min sharp.

#### Students do

- Write 1 notice + 1 wonder.
- Write one sentence about what they would do first to solve a radical equation.

#### Questions to ask

- (None during the phase — teacher harvests answers in Launch.)

**Adult role**

Solo teacher: circulate silently. Resist the urge to give hints.

**Do Now (5-4-savvas-model-discuss-lesson-5-4-launch)****EXPLORE & REASON**

- A. Solve  $3(a + 1)^2 + 2 = 11$ . Use at least two different methods.
- B. Try each of the methods you used in part (a) to solve  $3\sqrt{a + 1} + 2 = 11$ .
- C. Generalize Which of the methods is better suited for solving an equation with a radical? What problems arise when using the other method?

**[CHECK] ANTICIPATED STUDENT RESPONSES**

NOTICE (typical): “There’s a square root.” “It’s not a normal equation.” “The variable is under a radical sign.”

WONDER (typical): “How do you undo a square root?” “Do you square both sides?” “Could there be no solution?”

SENTENCES: students might write “square both sides” or “isolate the radical first.”

KEY MOVE in Launch: surface that squaring both sides is a one-way operation — it can create solutions that don’t work in the original. That’s why CHECK is non-negotiable.

*Tag: teacher models Example 1 + Example 3*

**Launch****DOK: 1-2****Minutes: 12****Teacher does**

- Bridge from Do Now: “You noticed the square root. To undo it, we square both sides — but that move can create fake solutions, so we always check.”
- Project Example 1 (Savvas Topic 5-4). Think aloud: isolate the radical, square both sides, solve, check the candidate in the ORIGINAL equation.
- Project Example 3. Think aloud: isolate, square, solve, check — one candidate fails the check (extraneous). Cross it out.
- Pause 1× for 30-sec partner-check: “In Example 1, what step would catch a fake solution?”
- Do NOT solve Try-Its. Release promptly at minute 17.

**Students do**

- Watch silently through Example 1.
- During partner-check: identify the check step.
- Copy the Isolate–Raise–Check Rules card into their page margin.
- Note the COMMON ERROR warning: always substitute into the ORIGINAL, not the squared form.

**Questions to ask**

- “What must I do before squaring?”
- “After squaring, how do I verify my answer?”
- “In Example 3, why is one solution extraneous?”

**Adult role**

Project, think aloud. Do NOT give students Try-Its to work until minute 17.

**[MIC] TEACHER SCRIPT**

1. “From the Do Now, you noticed the radical. The first move is always: isolate the radical on one side.”
2. Example 1 walk-through: “Watch me. I isolate the radical, then square both sides. Now I have a polynomial equation I can solve normally. Then I CHECK my answer in the ORIGINAL — not the squared form.”
3. Example 3 walk-through: “This time I get two candidates. I check both. One works; one does NOT. That failing one is extraneous — I cross it out and write ‘extraneous’ next to it.”
4. Common-error flag: “Never skip the check. Squaring is not reversible — it can create solutions that never existed in the original equation.”
5. Release: “Now you try. Try-It 1 asks you to isolate and solve. Try-It 3 asks you to find and reject an extraneous solution.”

*Tag: Think-Pair-Share + Practice*

**Explore**

**DOK: 1-2**

**Minutes: 33**

**Teacher does**

- 5 laps circulation across 33 min (~6–7 min each).
- Lap 1: Try-It 1 — press pairs to isolate first before squaring.
- Lap 2: Try-It 3 — press pairs to show the check step explicitly.
- Lap 3: #22 + #8 — check that pairs are substituting into the ORIGINAL equation.
- Lap 4: #15 — verify pairs identify and label the extraneous solution.
- Lap 5: #4 (extension) — stretch. Cut if pace slows.
- Do not give answers. Use sentence frame to press justification.

**Students do**

- Pairs move in order through 6 items: 2 Try-Its + 4 Practice.
- For each item: isolate the radical FIRST, then raise, then check.
- For Practice #15 (identify extraneous): show the check step and cross out the extraneous solution.
- Justify with sentence frame before writing.

**Questions to ask**

- Did you isolate the radical before raising both sides?
- Which equation did you substitute into to check — original or squared?
- If a candidate doesn’t satisfy the original, what do you write?
- Can you use the sentence frame to explain your answer?
- For #4: show all steps including the check.

**Adult role**

Solo teacher: 5 laps. Press pairs to justify using the rule card. No answers.

**Try It 1**

Solve each radical equation. Check your solution by substituting into the given equation or graphing.

a.  $\sqrt{x-2} + 3 = 5$

b.  $\sqrt[3]{x-1} = 2$

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (Savvas TE): A.  $x = 6$ ; B.  $x = 9$ .

**Try It 3 — extraneous solution**

Solve each radical equation. Check for any extraneous solutions.

a.  $x = \sqrt{7x+8}$

b.  $x + 2 = \sqrt{x+2}$

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (Savvas TE): A.  $x = 8$ ;  $x = -1$  is an extraneous solution. B.  $x = -1$ ,  $x = -2$ .

**Practice #22 — Solve radical equation**

Solve each radical equation.  $\sqrt{4x} = 11$

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from source):  $x = 30.25$ .

**Practice #8 — Solve radical equation**

$\sqrt{8x+9} = x$

**[CHECK] ANSWER / ANTICIPATED TALK**

Savvas TE answer:  $x = -1$ . Teacher note: squaring gives candidates  $x = 9$  and  $x = -1$ ;  $x = -1$  is extraneous, so the valid solution to the original equation is  $x = 9$ .

**Practice #15 — Identify extraneous**

Generalize Explain how to identify an extraneous solution for an equation containing a radical expression.

**[CHECK] ANSWER / ANTICIPATED TALK**

Substitute each potential solution into the original equation. If it makes the equation true, it is a real solution.

**Practice #4 — Solve and verify**

Error Analysis Nazim said that  $-3$  and  $6$  are the solutions to  $\sqrt{3x+18} = x$ . What error did Nazim make?

**[CHECK] ANSWER / ANTICIPATED TALK**

Nazim didn't check for extraneous solutions. The real solution is  $6$ ;  $(-3)$  is extraneous.

Tag: academic conversation

**Teacher does**

- Cold-call 2 pairs to read their sentence-frame sentence.
- Write one student-generated sentence on board.
- Preview P2: “Next class we’ll tackle equations with two radicals or a radical and a binomial — the same Isolate–Raise–Check logic applies.”

**Students do**

- Presenting pair reads their sentence.
- Class signals thumbs up/sideways.
- Note: next class extends this to more complex radical equations.

**Questions to ask**

- What’s a sentence that captures today’s rule about extraneous solutions?
- Why can’t we trust a candidate solution without checking it?

**Adult role**

Cold-call 2 pairs. Keep it brief — 5 min.

Pair share: “I isolated the radical to get \_\_\_\_\_. Squaring both sides gives \_\_\_\_\_. My candidate solutions are \_\_\_\_\_. When I check in the ORIGINAL equation, \_\_\_\_\_ is valid and \_\_\_\_\_ is extraneous because \_\_\_\_\_.”

Whole class: one pair reads their sentence. Class signals thumbs up/sideways.

*Tag: summary recap (not graded)*

**Exit Ticket****DOK: 1-2****Minutes: 2****Teacher does**

- Collect at door.
- Scan the fill-in items — misconceptions about the check step or extraneous solutions tell you what to re-hit in P2 Do Now.

**Students do**

- Complete the fill-in stems.
- Be honest about confusion.

**Questions to ask**

- Did students correctly identify what an extraneous solution is and why it arises?

**Adult role**

Collect. No grade.

### [CLIPBOARD] EXIT TICKET — Student Form

To solve a radical equation I first \_\_\_\_.

I raise both sides to power \_\_\_\_ because \_\_\_\_.

An extraneous solution happens when \_\_\_\_.

One biggest thing I learned today: \_\_\_\_\_

### OPTIONAL REINFORCEMENT (not collected)

*If you finish Explore early or want extra practice before Period 2.*

### [SEARCH] OPTIONAL • Model & Discuss

Return to the Do Now equation. With your partner:

B) Use Structure. Why can squaring both sides introduce a solution that doesn't work in the original equation?

C) Examine at least two equations you solved in Explore. For each, explain in your own words why checking is a required step and not optional.

#### Optional #1 (Savvas Practice)

Solve each radical equation.  $\sqrt[3]{x} + 8 = 13$

### [CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source):  $x = 125$ .

### [CLOCK] PERIOD A & F — 65-MIN CLASS NOTES

Both sections get 65 min for P1. Use the extra 10 min on Explore, NOT Launch.

Suggested add: project Savvas Practice #15 and #4 on screen for 2 more TPS rounds if pairs finish fast.

Or: extend Model & Discuss Parts B + C with “write one more radical equation that has an extraneous solution — then solve and reject it.”

### [PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Isolate–Raise–Check Rules card as a sticker/cut-out on their desk.
- For Try-It 1, allow students to use a step-by-step graphic organizer (isolate → raise → check).
- Accept verbal sentence-frame completions in lieu of written.

### [GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames printed on student packet (don’t skip).
- Word card: “extraneous solution” = a candidate answer that does not satisfy the original equation; must be rejected.
- “isolate” = get the radical alone on one side of the equation.
- “raise to the index power” = square both sides for a square root; cube both sides for a cube root.
- Pair ELL students with English-dominant partners for TPS.